

TLS_to_FDS: Technical Documentation and User Manual

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Summary

TLS_to_FDS is an open-source software framework and desktop Graphical User Interface (GUI) engineered to convert three-dimensional (3D) Terrestrial Laser Scanning (TLS) point clouds into fully formatted, physics-based Fire Dynamics Simulator (FDS) computational domains.

Numerical fire spread models such as FDS require spatially explicit, discretized representations of forest fuel beds (voxel matrices) along with thermodynamic, aerodynamic, and environmental boundary parameters. Manual creation of these inputs from raw or semantically classified remote sensing data is technically challenging, time-consuming, and prone to error.

TLS_to_FDS automates this workflow by providing: 1. **C-Accelerated Voxelization**: Fast spatial indexing and point-density-driven bulk density scaling. 2. **Dynamic 1D Boundary Fuel Model (BFM) Litter Discretization**: Two decoupled scientific models (Stem Distance Decay and Canopy Turnover Integration with Gaussian Wind Dispersion) discretized into 1D BFM &SURF classes and contiguous 2D ground &VENT patch tiles. 3. **Atmospheric & Firebrand Physics**: Atmosphere and wind field setup, dynamic ignition footprint generation, and Lagrangian ember transport with empirical char/ash material density overrides. 4. **Automated Multi-Mesh MPI Decomposition**: Parallel domain partitioning, coarse sky mesh placement, and automatic diagnostic output setup (Rate of Spread trackers, multi-height slice planes, and restart checkpoints). 5. **Vectorized BDF Binary Exporter**: High-performance structured Fortran Binary Data (.bdf) file generator.

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1. General Use and Workflow

TLS_to_FDS features an intuitive desktop interface built with PySide6 (Qt6). The application follows a 7-step sequential workflow:

TLS_to_FDS GUI WORKFLOW		
1. Select Input Directory	--->	Folder containing segmented .las / .laz files
2. Select Output Directory	--->	Destination for .fds and .bdf binary outputs
3. Select Forest Preset	--->	Biome default properties (Ponderosa, Oak, etc.)
4. Configure Fuel Layers	--->	Assign point cloud files to fuel classes
5. Set Spatial & Mesh	--->	Resolution (dx), domain padding, MPI partitions
6. Set Wind & Ignition	--->	Speed, direction, ABL stability, ignition pattern
7. Generate FDS Domain	--->	Assembles .fds master file and .bdf matrices

Core Project Parameters

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
Input Directory	line_input_dir	<i>Empty</i>	Path to the directory containing pre-segmented or unsegmented point cloud files (.las, .laz, .ply, .xyz).
Output Directory	line_output_dir	<i>Empty</i>	Destination folder where the master FDS script (.fds) and binary Fortran voxel matrices (.bdf) are saved.

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
Forest Preset	combo_preset	Ponderosa Pine Summer	Biome preset selector that automatically populates literature-derived combustion properties (Bulk Density, Moisture, Surface-to-Volume ratio, Element length, Drag) for each semantic vegetation layer.

2. Input Files and Point Cloud Processing

TLS_to_FDS ingests 3D point cloud datasets containing spatial coordinates (X, Y, Z) and optional classification values.

Input Modes

1. **Pre-Segmented Files:** The input directory contains separate `.las/.laz` files corresponding to individual vegetation strata (e.g. `grass.las`, `shrubs.las`, `stems.las`, `canopy.las`).
2. **UPCOMING: Deep Learning Fallback Segmentation:** For single, unclassified point cloud files, TLS_to_FDS will incorporate an integrated deep learning semantic segmentation plugin (3DFoS; Janvier et al., 2026) based on the Point Transformer V3 architecture (Wu et al., 2023), automatically partitioning raw point clouds into ground, understory, stem, and canopy layers.

3. Fuel Settings

Wildland fuels are heterogeneous, porous media whose physical and thermal characteristics dictate ignition probability, pyrolytic gas evolution, and fire spread. TLS_to_FDS converts 3D discrete point clouds into continuum porous media and surface boundary conditions in FDS by explicitly parameterizing fundamental **fuel properties**:

1. **Fuel Composition & Organic Content:** Chemical makeup (cellulose, hemicellulose, lignin) and dry organic mass vs. non-combustible mineral ash content, specified via FDS material definitions (`MATL_ID`).
2. **Moisture Content:** Dry-weight fuel moisture fraction ($M_f = m_{\text{water}}/m_{\text{dry}}$), dictating the thermal energy absorbed as latent heat during water vaporization prior to pyrolysis.
3. **Bulk Density:** Dry fuel mass per unit volume (ρ_b , kg/m^3), defining the dry combustible mass packed into each 3D vegetative voxel or ground boundary fuel cell.
4. **Fuel-Element Size & Shape:** Characteristic particle length (L , m) and **fuel-element shape (characterized by surface-to-volume ratio, $S/V = A/V$, m^{-1})**. High S/V fine fuels (grasses, foliage) exhibit rapid radiative absorption and fast surface pyrolysis, whereas low S/V coarse fuels (stems, trunks) burn slower with sustained charring.
5. **Porosity & Packing Ratio:** Derived aerodynamic indicators calculated directly from voxel bulk density ($\beta = \rho_b/\rho_p$ and void fraction $\phi = 1 - \beta$, assuming solid particle density $\rho_p \approx 500 \text{ kg}/\text{m}^3$), governing oxygen entrainment into the fuel bed, radiative attenuation, and drag.
6. **Unevenness in Fuel Distribution & Spatial Continuity:** Intra-canopy spatial heterogeneity, localized gaps, stem-centered decay gradients, and vertical/horizontal **continuity of fuel elements** (e.g., canopy fuel bridges connecting surface beds to overhead crowns).

3.1 Vegetation Layer Parameters

Individual 3D point cloud strata (e.g., surface fuels, canopy fuels, tree stems) are loaded into the interactive **Fuel Layers Table** in the GUI. When a fuel class is assigned to a layer, its physical, thermal, and aerodynamic combustion properties are automatically populated from the selected **Forest Preset** JSON configuration file (e.g., *Ponderosa Pine Summer*, *European Mixed Forest*, etc.). Users can also manually customize or override any of these values directly in the table:

Parameter	GUI Name	Default Value	Description, Logic & Scientific Basis
Nominal Bulk Density	bd	Layer dependent (0.5 – 25.0 kg/m ³)	Dry fuel mass per unit volume (kg/m ³). Represents overall bulk density and packing ratio ($\beta = \rho_b / \rho_p$) of the fuel layers. (<i>Ref: Rothermel, 1972</i>).
Moisture Content	moisture	Fraction (0.10 – 0.30)	Dry-weight moisture fraction $M_f = \frac{m_{\text{water}}}{m_{\text{dry}}}$. Controls latent heat of vaporization during thermal degradation. (<i>Ref: Van Wagner, 1977</i>).
Surface-to-Volume Ratio	sv_ratio	500 – 8000 m ⁻¹	Governs fuel-element shape (characterized by surface to volume ratio, $S/V = A/V$) . Controls radiative absorption, pyrolytic gas evolution, and surface area exposed to oxygen. High S/V (grass) ignites faster than low S/V (stems). (<i>Ref: Catchpole & de Mestre, 1986</i>).
Particle Length	length	0.10 – 0.50 m	Characteristic fuel-element size (L) . Used in FDS drag and radiative transfer formulations for particles.
Drag Coefficient	drag	2.8	Aerodynamic drag coefficient (C_d) of porous vegetative obstacles. Governed by fuel porosity, fuel-element shape and particle length. (<i>Ref: Mell et al., 2007; FDS Tech Guide</i>).

Parameter	GUI Name	Default Value	Description, Logic & Scientific Basis
Dynamic BD	<code>dynamic_bd</code>	True	Toggles 2-stage micro-voxelization dynamic bulk density imputation. Captures the unevenness in fuel distribution and spatial continuity of fuel elements based on local micro-voxel occupancy (1 cm) while preserving total dry fuel mass.

2-Stage Voxelization Dynamic Bulk Density Imputation To capture intra-canopy foliage clustering, gaps, and crown architecture while eliminating point-density bias from multi-scan coregistration overlaps, beam divergence, and range attenuation, TLS_to_FDS implements an **experimental 2-stage voxelization algorithm**:

1. **Stage 1 (1 cm Micro-Voxelization)**: The raw 3D point cloud of the vegetation layer is pre-voxelized at a fine sub-voxel resolution (`sub_voxel_size` = 0.01 m = 1 cm) using C-accelerated spatial hashing (**dendroptimized**). This collapses redundant or overlapping LiDAR returns occurring within the same 1 cm³ spatial volume into a single active micro-voxel center coordinate, decoupling point counts from multi-scan alignment inflation.
2. **Stage 2 (Simulation Voxelization & Occupancy Count P_v)**: The unique 1 cm micro-voxel centers are then voxelized at the target simulation grid resolution (Δx = `voxel_size`, e.g. 0.25 m). The number of occupied 1 cm micro-voxels within simulation voxel v is recorded as P_v .
3. **Stage 3 (Mean Occupancy & Density Correction)**: The mean micro-voxel occupancy across all non-empty simulation voxels in that fuel layer (N_{fl}) is computed:

$$\bar{P}_{fl} = \frac{1}{N_{fl}} \sum_{v=1}^{N_{fl}} P_v$$

The unconstrained voxel bulk density is scaled proportionally:

$$BD_v^{\text{raw}} = BD_{\text{nominal}} \cdot \left(\frac{P_v}{\bar{P}_{fl}} \right)$$

4. **Stage 4 (Safety Bounds & Mass Preservation)**: To prevent extreme density spikes from scan artifacts or sparse outliers, safety ratio bounds are applied:

$$BD_v^{\text{clamped}} = \max(0.05 \cdot BD_{\text{nominal}}, \min(4.0 \cdot BD_{\text{nominal}}, BD_v^{\text{raw}}))$$

Finally, the bulk density values are re-normalized to guarantee exact preservation of total dry fuel mass ($\sum BD_v \cdot V_{\text{voxel}} = M_{\text{nominal}}$):

$$BD_v = BD_v^{\text{clamped}} \cdot \left(\frac{N_{fl} \cdot BD_{\text{nominal}}}{\sum_{v=1}^{N_{fl}} BD_v^{\text{clamped}}} \right)$$

Vectorized Fortran Binary Data Format (.bdf) Export Architecture In FDS, vegetative fuel matrices are read as unformatted Fortran sequential binary streams (**.bdf**). Each file contains: 1. **Header Records**: - **Record 1 (48 bytes)**: Computational bounding box [$X_{\min}, X_{\max}, Y_{\min}, Y_{\max}, Z_{\min}, Z_{\max}$] as 6

double-precision floats (float64). - **Record 2 (24 bytes)**: Voxel cell dimensions $[\Delta x, \Delta y, \Delta z]$ as 3 double-precision floats (float64). - **Record 3 (4 bytes)**: Total active voxel count N as a 32-bit integer (int32). 2. **Body Records (Per-Voxel Stream)**: - **Record $2i + 4$ (24 bytes)**: Voxel centroid coordinate vector (x_i, y_i, z_i) (float64). - **Record $2i + 5$ (8 bytes)**: Local bulk density ρ_i (float64).

Standard Fortran unformatted I/O wraps each logical record with 4-byte unsigned integer markers (uint32) denoting the record's byte length. In naive implementations, writing $N = 500,000$ voxels via per-element Python loops requires 1,000,000 individual function calls, 1,000,000 transient NumPy allocations, and hundreds of thousands of individual filesystem writes (taking ≈ 45 seconds).

TLS_to_FDS eliminates this bottleneck using a **Zero-Copy Structured NumPy Interleaving Architecture**:

```
dtype_voxel = np.dtype([
    ("tag_c1", "<u4"),      # 24 (4 bytes)
    ("coords", "<f8", (3,)), # (X, Y, Z) centroid (24 bytes)
    ("tag_c2", "<u4"),      # 24 (4 bytes)
    ("tag_b1", "<u4"),      # 8 (4 bytes)
    ("bd", "<f8"),          # Bulk density (8 bytes)
    ("tag_b2", "<u4"),      # 8 (4 bytes)
])
```

The entire per-voxel payload (48 bytes/voxel) is generated in contiguous memory using vectorized array assignments and flushed to disk in a **single system write operation (f.write())**. This achieves a $\approx 1,000\times$ **speedup** (sub-50 ms for 500,000+ voxels) with **100% byte-for-byte Fortran binary compliance**.

3.2 Ground Fuels (Litter and Duff)

Definitions & Physical Characteristics

- **Surface Litter Bed**: The upper, un-compacted ground fuel layer consisting of freshly deposited dead plant material, including conifer needles, broadleaf foliage, twigs, small branches (≤ 6 mm diameter), and reproductive structures (cones, seed pods). Litter beds are physically characterized by high porosity, low dry bulk density ($\rho_{\text{litter}} \approx 10 - 25 \text{ kg/m}^3$), high surface-to-volume ratio ($S/V \approx 4000 - 9000 \text{ m}^{-1}$), and rapid moisture exchange with atmospheric relative humidity.
- **FUTURE UPDATE: Duff Bed**: The underlying organic layer extending from the base of the litter bed down to mineral soil, composed of partially decomposed (fermentation layer) and highly decomposed (humus layer) organic material. Duff beds are characterized by higher bulk density ($\rho_{\text{duff}} \approx 40 - 150 \text{ kg/m}^3$), lower S/V ratio, high mineral ash content, and high water holding capacity.

Numerical Representation: 1D Boundary Fuel Model (BFM) Grid Tiles Modeling thin surface fuel beds ($h \approx 0.01 - 0.05$ m) directly in 3D CFD grids ($\Delta z \approx 0.25 - 0.50$ m) creates severe numerical dilution: assigning a 5 cm litter bed of 15 kg/m^3 to a 50 cm voxel reduces effective 3D bulk density tenfold ($\rho_{3D} = 1.5 \text{ kg/m}^3$), causing under-predicted surface flame spread and artificial extinction.

TLS_to_FDS solves this by representing ground fuels as **1D Boundary Fuel Models (BFM)** mapped to 2D &VENT surface patches on the ground mesh boundary ($Z = 0$), utilizing multi-layer solid heat conduction and pyrolysis kinetics (&SURF MATL_ID(1:2, 1) with GENERIC VEGETATION over SOIL).

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
Enable Litter Layer	check_litter_active	False	Toggles inclusion of a ground litter layer.

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
Litter Model Mode	combo_litter_mode	Uniform	Chooses between Uniform (homogeneous load), Model 1: Tree Distance (stem decay), or Model 2: Canopy Turnover (vertical integration).
Litter Depth	spin_litter_depth	0.05 m	Physical thickness of the surface litter bed (h). (<i>Ref: Albini, 1976</i>).
Litter Bulk Density	spin_litter_bd	15.0 kg/m ³	Baseline dry bulk density of the ground fuel bed.
Litter Moisture	spin_litter_moisture	0.10 (10%)	Surface fuel moisture fraction.
Number of Litter Bins	spin_num_litter_bins	10	Number of discrete bulk density classes used to bin continuous spatial litter density maps into FDS &SURF definitions (Litter_Class_1 through Litter_Class_N) and row-merged &VENT patches.
Tree Map Path	line_tree_map	<i>Empty</i>	Path to CSV/TXT file containing (X, Y) stem coordinates for Model 1.
Radial Decay Rate (α)	spin_decay_alpha	0.5	Exponential decay coefficient in Model 1: $\rho(r) = \rho_{\min} + (\rho_0 - \rho_{\min})e^{-\alpha r},$ controlling how rapidly litter density drops with distance r from stems.
Min Litter Bulk Density	spin_min_litter_bd	2.0 kg/m ³	Background litter bulk density far from tree trunks in Model 1.
Canopy Turnover Rate	spin_turnover_rate	0.20 (20%/yr)	Annual foliage/branch drop fraction shed from overhead canopy in Model 2. (<i>Ref: Keane, 2015</i>).
Accumulation Time	spin_accumulation_years	3.0 years	Years of unburned litter accumulation or time since fire (t) in Model 2.

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
Decomposition Rate (k)	spin_decomposition_rate	0.15 yr ⁻¹	Annual exponential decomposition decay coefficient (k) in Model 2 based on the Olson accumulation model. Prevents infinite accumulation by converging to steady-state carrying capacity L/k . (Ref: Olson, 1963; Barber & Van Lear, 1984; Sánchez-López et al., 2026).
Fire Consumption (C)	spin_consumption_rate	1.0 (100%)	Fraction of available surface litter consumed during prior fire ($0 \leq C \leq 1$). When $C < 1.0$, accounts for steady-state unburnt residual fuel carry-over from recurring fire return intervals. (Ref: Sánchez-López et al., 2026).
Dispersion Sigma (σ)	spin_dispersion_sigma	1.5 m	Standard deviation (m) of the isotropic (omni-directional) 2D Gaussian kernel modeling multi-seasonal turbulent wind dispersion of falling canopy foliage. Guarantees mass conservation via normalization. (Ref: Mell et al., 2026; McDanold et al., 2023 [DUET]; Sánchez-López et al., 2026).
DTM File Path	line_dtm_path	Empty	Path to Digital Terrain Model point cloud or raster used for ground surface elevation referencing.

Scientific Ground Fuel Accumulation Models

1. Uniform Litter Model Applies a homogeneous bulk density (ρ_{litter}) and thickness (h) across the entire forest ground plane. Recommended for open grasslands, shrublands, or uniform plantation stands where spatial litter variations are negligible.

2. Model 1: Tree Distance Decay Model Models localized litter accumulation concentrated around tree trunks due to needle cast, branch fall, and bark shedding. Bulk density $\rho(x, y)$ decays exponentially as a function of Euclidean distance $d_k(x, y) = \sqrt{(x - x_k)^2 + (y - y_k)^2}$ to all tree stem locations (x_k, y_k) :

$$\rho(x, y) = \rho_{\min} + (\rho_0 - \rho_{\min}) \cdot \left[1 - \exp \left(- \sum_{k=1}^{N_{\text{stems}}} e^{-\alpha d_k(x, y)} \right) \right]$$

where ρ_0 is the peak nominal litter bulk density near tree stem locations (`spin_litter_bd`, e.g. 15.0 kg/m³), $\rho_{\min}$ is the background inter-stem litter bulk density far from tree trunks (`spin_min_litter_bd`, e.g. 2.0 kg/m³), and α is the radial spatial decay coefficient (1/m).

3. Model 2: Canopy Turnover, Annual Decomposition & Fall Dispersion Model Calculates spatially explicit litter mass load (W_{litter} , kg/m²) directly from overhead 3D canopy LiDAR point clouds and process-based ecological accumulation: 1. **Vertical Canopy Integration:** Integrates point-density corrected voxel bulk densities vertically to compute Canopy Fuel Load (CFL, kg/m²):

$$\text{CFL}(x, y) = \sum_{k=1}^{N_z} \rho_{\text{canopy}}(x, y, z_k) \cdot \Delta z$$

2. **Annual Deposition Rate:** Computes annual canopy litterfall production rate $L(x, y)$ (kg m⁻²yr⁻¹):

$$L(x, y) = \text{CFL}(x, y) \cdot k_{\text{turnover}}$$

3. **Olson Annual Decomposition & Fire Consumption Dynamics:** Applies the non-zero initial condition Olson fuel accumulation model (*Sánchez-López et al., 2026, Fire Ecology, Eq. 4*):

$$W_{\text{direct}}(x, y) = \frac{L(x, y)}{k} (1 - e^{-kt}) + (1 - C)B^{ss}(x, y)e^{-kt}$$

where k is the annual decomposition rate (`spin_decomposition_rate`, yr⁻¹), t is the accumulation period (`spin_accumulation_years`, yr), C is the fire consumption fraction (`spin_consumption_rate`, $0 \leq C \leq 1$), and $B^{ss}(x, y)$ is the pre-fire steady-state fuel load under recurring fire return intervals (FRI = t):

$$B^{ss}(x, y) = \frac{L(x, y)}{k} \cdot \frac{1 - e^{-kt}}{1 - (1 - C)e^{-kt}}$$

(Note: When $C = 1.0$ (complete consumption), this reduces to the classic single-cycle Olson (1963) model: $W(x, y) = \frac{L(x, y)}{k}(1 - e^{-kt})$. In the limit $k \rightarrow 0$, it smoothly recovers the linear accumulation model $L(x, y) \cdot t$.)

4. **Isotropic 2D Gaussian Wind Dispersion:** Models turbulent, multi-seasonal, omni-directional wind advection and dispersion of falling foliage during drop via a radially symmetric (isotropic) 2D Gaussian convolution:

$$W_{\text{dispersed}}(x, y) = \iint W_{\text{direct}}(x', y') \cdot \frac{1}{2\pi\sigma^2} \exp \left(-\frac{(x - x')^2 + (y - y')^2}{2\sigma^2} \right) dx' dy'$$

The dispersed load matrix is re-scaled to guarantee exact mass conservation ($\sum W_{\text{dispersed}} = \sum W_{\text{direct}}$) and clamped strictly to the forest footprint boundary. (Note: This formulation assumes isotropic (omni-directional) dispersion, reflecting the time-integrated effect of variable seasonal wind gusts and turbulence over the multi-year accumulation period rather than a single unidirectional wind vector. This builds upon the mechanistic canopy litterfall dispersion principles established in models like DUET (McDanold et al., 2023) and empirical isotropic kernel smoothing (Sánchez-López et al., 2023, 2026)).

Discrete Bounding Box Voxel/Vent Discretization Continuous 2D spatial bulk density grids $\rho(x, y)$ are discretized into N_{bins} discrete SURF classes. Contiguous cells sharing identical SURF classes are coalesced into rectangular 2D FDS & VENT patches via 2D scanline run coalescing, reducing total namelist entries in .fds files by up to 90% and accelerating CFD initialization.

Ground Fuel Spatial Raster Exports When ground litter layers are generated (using Uniform, Model 1, or Model 2), TLS_to_FDS automatically exports 2D spatial raster grids, matrices, and visual heatmaps to the destination output folder for GIS analysis, numerical post-processing, and inspection: - **litter_bulk_density.tif**: Standard GeoTIFF Raster Grid of 2D ground litter bulk density (kg/m^3). Features georeferenced spatial dimensions and native drag-and-drop compatibility with QGIS, ArcGIS, GDAL, and rasterio. - **litter_fuel_load.tif**: Standard GeoTIFF Raster Grid of 2D surface fuel load ($W = \rho_{\text{litter}} \cdot h_{\text{litter}}$, kg/m^2). - **litter_bulk_density.png**: High-resolution 2D visual heatmap preview displaying spatial axes in meters, colormap in kg/m^3 , statistical metrics box (mean, min, max, std), and tree stem overlay coordinates (in Model 1). - **litter_fuel_load.png**: High-resolution 2D visual heatmap preview displaying surface fuel load in kg/m^2 . - **litter_bulk_density.csv**: Plain-text CSV matrix of bulk density values (kg/m^3) for custom numerical scripting. - **litter_fuel_load.csv**: Plain-text CSV matrix of surface fuel load values (kg/m^2) for custom numerical scripting.

4. Fire and Wind Settings

The **Fire and Wind Settings** module parameterizes atmospheric inflow dynamics, boundary layer velocity profiles, burner ignition geometry, and Lagrangian ember lofting kinetics.

4.1 Wind Field & Atmospheric Forcing

Physical & Numerical Basis Wind forcing drives convective heat transfer ahead of the fire front, tilts flame structures, and accelerates rate of spread (RoS). In TLS_to_FDS, atmospheric wind is initialized at computational domain boundaries using open mesh inflow/outflow conditions (&VENT MB='XMIN' /, &VENT MB='YMIN' /, etc.) paired with logarithmic boundary layer velocity profiles.

To prevent numerical shocks and ensure realistic fire-atmosphere coupling: - **Wind Development Time** (t_{dev}): Fire simulations require a wind stabilization period prior to burner ignition. During t_{dev} , the CFD solver computes wind flow through the vegetative canopy obstacles, establishing logarithmic boundary layer turbulence and wake eddies across the domain before ignition begins.

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
Wind Development Time	spin_wind_dev	25.0 s	Pre-ignition stabilization period (t_{dev}). Allows atmospheric boundary layer wind profiles to stabilize and traverse the domain before fire ignition commences. (Ref: <i>Simó-Martí et al., 2026</i>).
Wind Direction	spin_wind_dir	270°	Meteorological wind origin direction: 0° (North), 90° (East), 180° (South), 270° (West, flowing East). Controls inflow/outflow mesh vent assignment.

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
Wind Speed	<code>spin_wind_speed</code>	3.0 m/s	Reference wind speed (U_{ref}) applied at the top of the atmospheric boundary layer or reference anemometer height.

4.2 Atmospheric Boundary Layer Physics

Monin-Obukhov Similarity Theory Formulation The atmospheric boundary layer (ABL) velocity profile $u(z)$ above the ground and through the canopy drag layer is parameterized via Monin-Obukhov Similarity Theory (MOST):

$$u(z) = \frac{u_*}{\kappa} \left[\ln \left(\frac{z + z_0}{z_0} \right) - \psi_m \left(\frac{z}{L} \right) \right]$$

where u_* is friction velocity, $\kappa = 0.41$ is the von Kármán constant, z_0 is aerodynamic roughness length, L is the Obukhov length, and ψ_m is the universal stability correction function.

In neutral atmospheric conditions ($|L| \rightarrow \infty$ or $L = -500$ m), thermal stability effects vanish ($\psi_m = 0$), reducing the formulation to the classic logarithmic law of the wall:

$$u(z) = \frac{u_*}{\kappa} \ln \left(\frac{z + z_0}{z_0} \right)$$

This ensures that wind speed drops smoothly to zero at the surface roughness height (z_0), preventing numerical pressure shocks at domain boundaries and establishing realistic vertical wind shear before ignition starts.

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
Obukhov Length (L)	<code>spin_obukhov</code>	-500.0 m	Characterizes lower atmospheric thermal stability. $L < 0$ denotes unstable convective conditions (strong thermal updrafts); $L > 0$ denotes stable conditions; -500 m or ∞ represents neutral stratification. (<i>Ref: Monin & Obukhov, 1954; FDS User Guide</i>).

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
Aerodynamic Roughness (z_0)	spin_z0	0.5 m	Ground roughness length defining the height where logarithmic wind speed drops to zero. Standard Davenport-Wieringa terrain values: 0.03 m (open grass), 0.25 m (scattered trees), 0.5 m (forest stands), 1.0 m (closed forests). (<i>Ref: Wieringa, 1992</i>).

4.3 Ignition Boundary Setup

Thermal Pyrolysis Initiation & Supported Ignition Geometries Ignition is simulated using artificial surface burner vents (&VENT SURF_ID='IGN FIRE') located on the ground ($Z = 0$). The burner emits a specified Heat Release Rate Per Unit Area (HRRPUA, kW/m^2) for a finite duration (t_{ign}), radiating sufficient thermal energy to heat nearby vegetative voxels past their pyrolytic degradation threshold ($\approx 300^\circ\text{C}$).

TLS_to_FDS automatically calculates burner coordinates based on the unpadded forest bounding box footprint, supporting 8 standard ignition patterns:

- **Full-Width Line / Edge Fires:**
 - **Line: South Edge ($y_{\min}$):** Full-width ignition strip along the southern domain boundary ($Y_{\min}$).
 - **Line: North Edge ($y_{\max}$):** Full-width ignition strip along the northern domain boundary ($Y_{\max}$).
 - **Line: West Edge ($x_{\min}$):** Full-width ignition strip along the western domain boundary ($X_{\min}$).
 - **Line: East Edge ($x_{\max}$):** Full-width ignition strip along the eastern domain boundary ($X_{\max}$).
- **Corner Point Fires:**
 - **Point: South-West Corner:** Concentrated burner at ($X_{\min}, Y_{\min}$). Simulates simultaneous head-fire and flank-fire development under oblique wind angles.
 - **Point: South-East Corner:** Concentrated burner at ($X_{\max}, Y_{\min}$).
 - **Point: North-West Corner:** Concentrated burner at ($X_{\min}, Y_{\max}$).
 - **Point: North-East Corner:** Concentrated burner at ($X_{\max}, Y_{\max}$).

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
HRRPUA	spin_hrrpua	2500.0 kW/m^2	Heat Release Rate Per Unit Area for the ignition burner. Provides thermal energy required to initiate fuel pyrolysis. (<i>Ref: Babrauskas, 1984; Mell et al., 2007</i>).

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
Ignition Duration	spin_ign_duration	10.0 s	Time duration the artificial ignition line remains active before shutting off.
Ignition Vent Width	spin_vent_width	1.0 m	Physical width of the ignition vent strip along the domain boundary.
Ignition Pattern	combo_ign_pattern	Line: South Edge (y_min)	Geometric footprint of ignition: 4 edge line fires or 4 corner point burners. Automatically snaps to vegetation outer bounds at $Z = 0$.

4.4 Ember Transport and Firebrand Physics

Lagrangian Particle Tracking & Material Density Overrides Spotting caused by lofted firebrands is a primary driver of rapid wildfire propagation and crown fire acceleration. When enabled, TLS_to_FDS activates FDS Lagrangian particle tracking (EMBER_PARTICLE = .TRUE.):

1. **Char & Ash Density Transitions:** Thermal degradation of vegetative fuel elements produces char and ash residue. TLS_to_FDS automatically applies empirical material density overrides:

$$\rho_{\text{char}} = 125.0 \text{ kg/m}^3 \quad \text{and} \quad \rho_{\text{ash}} = 5.0 \text{ kg/m}^3$$

as established in empirical tree burning experiments (*Mell et al., 2026*).

2. **Ember Lofting Thresholds:** As vegetation burns and mass is lost, particles meeting the **Ember Density Threshold** ($\rho_{\text{ember}} \leq 62.5 \text{ kg/m}^3$) are converted into discrete particles. If localized convective plume updrafts exceed the **Ember Velocity Threshold** ($W_{\text{updraft}} \geq W_{\text{threshold}}$), the firebrands break away and are advected downwind by the turbulent flow field.

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
Enable Ember Transport	check_track_embers	False	Toggles Lagrangian firebrand particle generation. When active, TLS_to_FDS automatically applies empirical char density ($\rho_{\text{char}} = 180 \text{ kg/m}^3$) and ash density ($\rho_{\text{ash}} = 50 \text{ kg/m}^3$) material overrides. (<i>Ref: Mell et al., 2026</i>).
Ember Density Threshold	spin_ember_density	62.5 kg/m ³	Density threshold at which decaying vegetative particles convert into loftable char firebrands.

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
Ember Velocity Threshold	spin_ember_velocity	0.0 m/s	Critical localized updraft wind velocity required for aerodynamic lifting and firebrand breakaway into the plume flow. (<i>Ref: Manzello et al., 2008</i>).

5. Spatial Settings

The **Spatial Settings** module handles computational domain geometry, outer buffer padding, boundary snapping, coarse sky mesh generation, and parallel multi-core grid partitioning.

5.1 Domain Padding and Canopy Enclosure

Vertically Nested Multi-Grid Domain & Boundary Reflection Prevention To prevent artificial CFD boundary constriction and numerical reflection: 1. **Lateral Padding (X,Y)**: Lateral domain extensions ensure inflow wind profiles stabilize before striking fuel obstacles and outflow plumes exit without artificial pressure feedback. 2. **Top Sky Mesh (Z)**: Plume rise above dense canopies requires vertical clearance. TLS_to_FDS enforces the rule $Z_{\max} \geq 5 \times H_{\text{canopy}}$. To optimize cell counts, a **Vertically Nested Multi-Grid Domain** structure is used: a fine base mesh encloses the vegetation canopy ($\Delta x = \Delta y = \Delta z$), while a coarser sky mesh (e.g., $2\times$ or $3\times$ base cell size) covers the upper atmosphere.

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
X Domain Padding	spin_pad_x	10.0 m	Lateral domain extension along the X axis.
Y Domain Padding	spin_pad_y	10.0 m	Inflow and outflow domain extension along the Y axis.
Z Sky Padding	spin_pad_z	20.0 m	Vertical height of coarse sky mesh placed above the forest canopy. Enforces $Z_{\max} \geq 5 \times H_{\text{canopy}}$ to prevent top boundary flow constriction. (<i>Ref: McGrattan et al., 2013</i>).
Sky Mesh Multiplier	combo_sky_mult	2x	Grid coarsening factor for upper sky mesh (e.g. $2 \times \Delta x$).
Domain Geometry Wizard	btn_domain_wizard	N/A	Interactive tool that calculates optimal padding and multi-mesh grid dimensions based on target cell counts.

5.2 Parallel Multi-Mesh MPI Subdivisions

High-Performance MPI Domain Decomposition FDS uses Message Passing Interface (MPI) for domain decomposition, assigning individual 3D mesh blocks to separate CPU cores. TLS_to_FDS partitions the base domain along the X and Y axes into $\text{MPI}_X \times \text{MPI}_Y$ uniform sub-meshes, ensuring mesh boundaries align seamlessly with integer cell counts and coarse sky mesh boundaries.

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
MPI X Subdivisions	spin_mpi_x	2	Number of mesh domain partitions along the X axis (MPI_X).
MPI Y Subdivisions	spin_mpi_y	3	Number of mesh domain partitions along the Y axis (MPI_Y).

Total Computational Meshes = $(\text{MPI}_X \times \text{MPI}_Y) + 1_{\text{sky}}$ (e.g. $2 \times 3 = 6$ base meshes + 1 sky mesh = 7 parallel MPI processes).

6. Output Settings

The **Output Settings** module controls total simulation duration, diagnostic slice planes, boundary heat release rate maps, rate of spread trackers, and simulation restart checkpoints.

6.1 Simulation Timing and Volumetric Outputs

Simulation Duration & Diagnostic File Exports Located within the **Output Settings** tab in the GUI, users configure total run duration alongside 2D/3D diagnostic output controls:

- **Surface Heat Release Rate (&BNDF HRRPUA)**: Generates 2D maps of ground surface heat release rate to track fire front progress, flame residence time, and fireline intensity.
- **Volumetric HRR (&SLCF HRRPUV)**: Slices 3D heat release rate per unit volume (kW/m^3) to visualize flame geometry, tilt angle, and flame height.
- **Gas Temperature & Velocity Slices (&SLCF)**: Slices gas temperature ($^{\circ}\text{C}$) and velocity vectors (u, v, w) to analyze plume buoyancy, crown scorch risk, and wind-fire coupling.

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
Total Simulation Time	spin_sim_time	240.0 s	Total duration of the fire simulation in seconds (located in the Output Settings tab).
HRRPUA (Boundary File)	check_out_hrrpua	True	Exports 2D ground surface heat release rate maps (&BNDF). Essential for calculating Rate of Spread (RoS) and fireline intensity.

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
Volumetric HRR (Slice)	check_out_flame	True	Exports mid-plane 2D slices of volumetric heat release rate (&SLCF HRRPUV) to visualize 3D flame structure and flame height.
Gas Temperature (Slice)	check_out_temp	True	Exports mid-plane 2D gas temperature slices (&SLCF). Used for convective plume analysis and crown scorch modeling.
Wind Velocity (Slice)	check_out_wind	True	Exports 2D velocity vector slices (&SLCF U, W velocity) to analyze wind-fire coupling and fire-induced indrafts.
Dry Biomass Tracking	check_out_biomass	False	Tracks total dry fuel mass (kg) remaining in each vegetation stratum over time to determine percentage fuel consumption.

6.2 Rate of Spread (RoS) Tracking

Transverse Temperature Front Arrival Trackers To record exact fire front arrival times across the plot without manual post-processing, TLS to FDS can automatically insert 4 transverse device transect lines (&DEVC): - Devices measure MAXLOC X or MAXLOC Y temperature statistics at 20%, 40%, 60%, and 80% span across the domain. - Computes plot-scale Rate of Spread (m/s or m/min) directly from device output time series.

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
Enable RoS Trackers	check_ros_tracking	False	Generates 4 transverse device lines (&DEVC) measuring MAXLOC X or MAXLOC Y temperature statistics across domain span transects (at 20%, 40%, 60%, and 80% span). Automatically logs fire front arrival times across the plot.

6.3 Multi-Height Slice Planes

Vertical Profiling Through Canopy Strata Horizontal slice planes (&SLCF) can be requested at multiple vertical elevations Z :

GUI Parameter	UI Widget	Default Value	Description, Logic & Scientific Basis
Slice Heights (m)	<code>line_slice_heights</code>	"1.0"	Comma-separated list of horizontal elevation planes (e.g. 0.5, 1.5, 2.5) for multi-level &SLCF slice outputs. Enables vertical profiling of plume temperatures and horizontal wind speeds through understory and overstory canopy layers.

6.5 Interactive 3D Pre-Simulation Scene Visualization

Pre-Flight Simulation Domain Inspection Prior to executing computationally intensive CFD runs, `TLS_to_FDS` automatically generates an interactive 3D HTML visualization (`fds_scene_preview.html`) in the output destination folder. Built with Plotly 3D graphics, this lightweight, stand-alone file can be opened directly in any modern web browser without third-party plugins.

Visual Components & Physics Features

- Volumetric Voxel Cubes (`go.Mesh3d`):**
 - Voxels are rendered as 3D solid cubic meshes.
 - Voxel faces are color-mapped using the continuous `YlOrRd` (Yellow-Orange-Red) thermal intensity colormap based on per-voxel bulk density (kg/m^3).
 - Face-level hover tooltips display exact (X, Y, Z) centroid coordinates, cell dimensions, and bulk density.
- 1D BFM Ground Litter Surface Layer:**
 - The 2D surface litter bed is rendered at $Z = 0$ cropped to the unpadded forest footprint.
 - Interactive hover tooltips provide position (X, Y) , bulk density (kg/m^3), fuel load (kg/m^2), and physical layer depth (cm).
- Tree Stems & Ignition Footprints:**
 - In Model 1, tree stem coordinates are projected onto the ground plane as distinct 3D markers and coordinate pins.
 - The selected ignition boundary pattern (South/North/East/West edges, SW/SE/NW/NE corners, or custom spot fire) is rendered as a bright red burner vent polygon at $Z = 0$.
- 3D Inflow Wind Vector Arrows:**
 - Directional 3D wind velocity vector arrows (`go.Scatter3d`) are placed along the active domain boundary inflow face, visually confirming wind direction (degrees) and velocity magnitude (U_{ref}).
- Computational Domain & Coarse Sky Mesh Wireframes:**
 - Outer bounding boxes display the base domain and sky mesh boundaries, allowing visual verification of lateral domain padding and top sky clearance.

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8. Appendix A: Forest Biome Presets & Empirical Literature Reference

IN CONSTRUCTION

Please note that these presets are under development and will be updated TLS_to_FDS includes pre-configured, literature-derived combustion property presets for common forest biomes and species.

8.1 Preset Overview

Biome Preset	Semantic Layer	Bulk Density (kg/m ³)	Moisture (M_f)	S/V (m ⁻¹)	Length L (m)	Drag (C_d)	Embers
Ponderosa Pine (Summer)	Canopy Fuel	0.40	1.00 (100%)	3588	0.10	2.8	Yes
	Surface Fuel	0.80	0.15 (15%)	8000	0.10	2.8	Yes
	Ground Fuel	1.20	0.15 (15%)	9770	0.10	2.8	Yes

Biome Preset	Semantic Layer	Bulk Density (kg/m ³)	Moisture (M_f)	S/V (m ⁻¹)	Length L (m)	Drag (C_d)	Embers
Ponderosa Pine (Winter)	Trunks	10.00	1.00 (100%)	13	0.10	2.8	Yes
	Litter	15.00	0.05 (5%)	6000	0.10	2.8	Yes
	Canopy	0.40	1.20 (120%)	3588	0.10	2.8	Yes
	Fuel						
	Surface Fuel	0.80	0.35 (35%)	8000	0.10	2.8	Yes
	Ground Fuel	1.20	0.35 (35%)	9770	0.10	2.8	Yes
	Trunks	10.00	1.10 (110%)	13	0.10	2.8	Yes
European Mixed Forest (Sum- mer)	Litter	15.00	0.20 (20%)	6000	0.10	2.8	Yes
	Canopy	0.50	1.10 (110%)	4500	0.10	2.8	Yes
	Fuel						
	Surface Fuel	0.70	0.20 (20%)	7500	0.10	2.8	Yes
	Ground Fuel	1.00	0.20 (20%)	9000	0.10	2.8	Yes
	Trunks	12.00	1.00 (100%)	13	0.10	2.8	Yes
	Litter	20.00	0.12 (12%)	5500	0.10	2.8	Yes
European Mixed Forest (Winter)	Canopy	0.30	0.80 (80%)	3000	0.10	2.8	Yes
	Fuel						
	Surface Fuel	0.70	0.40 (40%)	7500	0.10	2.8	Yes
	Ground Fuel	1.00	0.40 (40%)	9000	0.10	2.8	Yes
	Trunks	12.00	1.10 (110%)	13	0.10	2.8	Yes
	Litter	20.00	0.30 (30%)	5500	0.10	2.8	Yes
	Canopy	0.40	0.80 (80%)	3588	0.10	2.8	Yes
Boreal Forest	Fuel						
	Surface Fuel	0.80	0.10 (10%)	8000	0.10	2.8	Yes
	Ground Fuel	1.20	0.10 (10%)	9770	0.10	2.8	Yes
	Trunks	10.00	0.90 (90%)	13	0.10	2.8	Yes
	Litter	15.00	0.05 (5%)	6000	0.10	2.8	Yes
	Canopy	0.24	1.10 (110%)	4500	0.10	2.8	Yes
	Fuel						
Mediterranean Stone Pine (<i>Pinus pinea</i>)							

Biome Preset	Semantic Layer	Bulk Density (kg/m^3)	Moisture (M_f)	S/V (m^{-1})	Length L (m)	Drag (C_d)	Embers
Mediterranean Maritime Pine (<i>Pinus pinaster</i>)	Surface	0.70	0.20 (20%)	7500	0.10	2.8	Yes
	Fuel						
	Ground	1.00	0.20 (20%)	9000	0.10	2.8	Yes
	Fuel						
	Trunks	12.00	1.00 (100%)	13	0.10	2.8	Yes
	Litter	20.00	0.12 (12%)	5500	0.10	2.8	Yes
	Canopy	0.20	1.10	6249	0.09	2.8	Yes
	Fuel		(110%)				
	Surface	0.70	0.20 (20%)	4850	0.006	2.8	Yes
	Fuel						
Mediterranean Aleppo Pine (<i>Pinus halepensis</i>)	Ground	17.04	0.07 (7%)	4892	0.09	2.8	Yes
	Fuel						
	Trunks	12.00	1.00 (100%)	13	0.10	2.8	Yes
	Litter	17.04	0.07 (7%)	4892	0.09	2.8	Yes
	Canopy	0.20	1.10	4892	0.18	2.8	Yes
	Fuel		(110%)				
	Surface	0.70	0.20 (20%)	2755	0.0035	2.8	Yes
	Fuel						
	Ground	17.55	0.118 (11.8%)	6249	0.10	2.8	Yes
	Fuel						
Blank / Custom	Trunks	12.00	1.00 (100%)	13	0.10	2.8	Yes
	Litter	17.55	0.118 (11.8%)	6249	0.10	2.8	Yes
Blank / Custom	All Layers	<i>User- Defined</i>	<i>User- Defined</i>	<i>User- Defined</i>	0.10	2.8	<i>Configurable</i>

8.1 Empirical Bulk Density Literature

IN CONSTRUCTION

Bulk density values (kg/m^3) reported across the wildfire and forestry literature vary substantially between **canopy bulk densities** (CBD) and **ground fuel beds** (ρ_{litter} , ρ_{duff} , ρ_{moss}):

1. Boreal Forest Ecosystems (*Picea mariana*, *Pinus sylvestris*, *Pinus banksiana*)

- *Canopy Bulk Density*: Typically ranges from 0.15 to 0.50 kg/m^3 across circumboreal stands. In dense black spruce (*Picea mariana*) stands, available crown fuel bulk density frequently reaches 0.40 to 0.70 kg/m^3 because of dense, club-like crown tops with semi-serotinous cones (*Alexander et*

al., 2004; Cruz *et al.*, 2005). In Scots pine (*Pinus sylvestris*), stands typically exhibit lower mean CBD (0.10 to 0.35 kg/m³).

- *Ground Moss Bed*: Living moss (*Pleurozium schreberi*, *Hylocomium splendens*) has an uncompacted bulk density of 15.0 to 25.0 kg/m³, exhibiting high porosity and rapid moisture exchange. Dead/decomposing lower moss layers reach 30.0 to 60.0 kg/m³ (Kasischke & Johnstone, 2005).
- *Boreal Duff Layer*:
 - *Pinus-Picea stands*: BD = 92.0 kg/m³, depth = 6.9 ± 0.12 cm (Miyanishi & Johnson, 2002).
 - *Picea stands*: BD = 94.0 kg/m³, depth = 10.3 ± 0.11 cm (Miyanishi & Johnson, 2002).

2. Eucalyptus Plantations (*Eucalyptus globulus*, *E. nitens*)

- *Canopy Bulk Density*: Ranges from 0.05 to 0.28 kg/m³. Dense, unthinned young *E. globulus* plantations in Spain and Portugal average 0.12 to 0.22 kg/m³, whereas mature or heavily thinned commercial stands exhibit values between 0.04 and 0.14 kg/m³ (Fernandes *et al.*, 2009; Gómez-Vázquez *et al.*, 2013).
- *Litter Layer*: Natural forest floor litter (composed of broad leaves, twigs, and decorticated ribbon bark) exhibits a mean bulk density of 23.6 ± 10.6 kg/m³ (ranging between 18.0 and 35.0 kg/m³) (Fernandes *et al.*, 2009).

3. Mediterranean Pine Ecosystems (*Pinus pinea*, *Pinus pinaster*, *Pinus halepensis*)

- *Stone Pine* (*Pinus pinea*): Characterized by an umbrella-shaped, spreading crown architecture.
 - *Canopy Bulk Density*: 0.07 to 0.30 kg/m³ (mean regional inventory values across Spain and Greece cluster around 0.14 to 0.18 kg/m³) (Mitsopoulos & Dimitrakopoulos, 2007; Molina *et al.*, 2019).
- *Maritime Pine* (*Pinus pinaster*): High resin content and long, thick needle cast.
 - *Canopy Bulk Density*: 0.10 to 0.32 kg/m³ (Álvarez-González *et al.*, 2010; Ruiz-González *et al.*, 2015).
 - *Litter Layer*: Needle litter beds typically range between 15.0 and 30.0 kg/m³.
- *Aleppo Pine* (*Pinus halepensis*): High canopy flammability with serotinous cone retention.
 - *Canopy Bulk Density*: 0.09 to 0.22 kg/m³ (Mitsopoulos & Dimitrakopoulos, 2007)
 - *Litter Layer*: Needle litter beds typically range between 15.0 and 28.0 kg/m³.
- *Mediterranean Sclerophyllous Shrub Understory* (*Cistus*, *Ulex*, *Erica*): -

9. Appendix B: Generated Output Files Summary

Upon clicking **Generate FDS Files**, TLS_to_FDS populates the selected output directory with simulation input scripts, binary matrices, rasters, and visual diagnostic tools:

Output File Name	File Format	Purpose & Compatible Software
<sim_name>.fds	FDS Input Script (.fds)	Master simulation file containing complete mesh, atmospheric boundary, burner, and diagnostic output namelists. Run directly via <code>fds <sim_name>.fds</code> .

Output File Name	File Format	Purpose & Compatible Software
<layer_name>.bdf	Fortran Binary Data (.bdf)	C-vectorized unformatted Fortran sequential binary voxel arrays containing (X, Y, Z) centroid coordinates and per-voxel bulk density (kg/m^3). Read automatically by FDS.
fds_scene_preview.html	Interactive 3D HTML Preview	Stand-alone 3D Plotly pre-simulation domain visualizer with cubic voxel meshes bulk-density-based colormaps, 3D wind vector arrows, tree stems, and burner footprints. Openable in any web browser.
litter_bulk_density.tif	GeoTIFF Raster (.tif)	Georeferenced 2D spatial raster grid of ground litter bulk density (kg/m^3). Native drag-and-drop support in QGIS, ArcGIS, GDAL, and Python rasterio.
litter_fuel_load.tif	GeoTIFF Raster (.tif)	Georeferenced 2D spatial raster grid of surface fuel load ($W = \rho_{\text{litter}} \cdot h$, kg/m^2).
litter_bulk_density.png	Publication PNG Heatmap	High-resolution 2D visual heatmap showing spatial axes in meters, colormap in kg/m^3 , statistical metrics box (mean, min, max, std), and tree stem coordinates (in Model 1).
litter_fuel_load.png	Publication PNG Heatmap	High-resolution 2D visual heatmap showing surface fuel load in kg/m^2 .
litter_bulk_density.csv	Numerical CSV Matrix	Plain-text numerical matrix of 2D bulk density values (kg/m^3) for custom mathematical scripting and spreadsheet analysis.
litter_fuel_load.csv	Numerical CSV Matrix	Plain-text numerical matrix of 2D surface fuel load values (kg/m^2).